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IT4010 Research Project – 2026 Jan

Topic Assessment Form

V2.1

Project ID: **R26-CS-010**

1. Topic (12 words max)

Privacy-Enhanced Synthetic Data Generation for Autonomous Vehicle Training

2. Research group the project belongs to

AIMS - Autonomous Intelligent Machines and Systems

3. Specialization of the project belongs to

Cyber Security (CS)

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (300 words max) – references not included in word count.

The systems for autonomous vehicles (AV) need to be supplied with extensive datasets in order to train the perception, prediction and decision-making parts of the system. Driving data from the real world usually comes with lots of sensitive personal information, such as people's faces, car license plates, location traces and behavioral patterns, thus posing major privacy and regulatory challenges. The parties involved mainly consist of manufacturers of AVs, AI researchers, regulators and the public whose data is collected. Limitations on the data use and sharing have a negative impact on the innovation, scalability and deployment of AV systems. The main users of the proposed solution are AV developers and research organizations that need the training data to be secure and meet the regulations.

This issue has broad impacts. Socially and ethically, privacy violations reduce public trust. Technically, limited data access lowers model robustness and safety. Economically, compliance requirements and legal risks increase development costs and slow progress. Legally, regulations such as GDPR impose strict rules on data collection, storage, and reuse. Environmentally, repeated real-world data collection consumes significant resources compared to simulation-based approaches.

The main issues concerning the use of real-world data for AV training are solved by the proposal of generating privacy-enhanced synthetic data. The method that is being used cuts down the use of sensitive real data and at the same time, keeps the utility of the model to be high by combining the application of privacy-preserving techniques with the data generation. The project not only complies with the AIMS research cluster but also with the Autonomous Vehicles and Intelligent Transportation Systems domain, with a cybersecurity focus. It helps the Sustainable

Development Goals by making secure innovation (SDG 9), safe and sustainable transport (SDG 11), and responsible data governance and privacy protection stronger and therefore, more difficult to challenge (SDG 16).

References

• European Union. General Data Protection Regulation (GDPR), 2016.
• Goodfellow, I. et al., "Generative Adversarial Networks," NeurIPS, 2014.
• Dosovitskiy, A. et al., "CARLA: An Open Urban Driving Simulator," CoRL, 2017.
• Dwork, C., "Differential Privacy," ICALP, 2006.

6. Brief description of Existing Research and Systems (300 words max)

The progress of self-driving cars (AVs) is supported by actual and simulated data for the training of machine learning algorithms. Datasets from real-world sources like KITTI, Waymo Open Dataset, Lyft Level 5, and nuScenes present marked sensor data for perception tasks such as object detection, tracking, and semantic segmentation. Nevertheless, the use of these datasets generates issues concerning privacy since they include sensitive data such as human faces, car plates, and location tracking, which are protected by rules like GDPR.

Synthetic data generation is concerned with the development of top-notch datasets of artificial origin by simulation-based algorithms like CARLA, AirSim, Unity3D, and NVIDIA Omniverse which can either support the original data or replace it entirely. The previously mentioned platforms can generate a variety of driving scenarios, changeable weather, and rare edge cases, and at the same time, significantly reduce the expenses of data collection. Besides, they allow automated labeling, faster iterations, and reduced operational costs to take place.

The academic world has investigated various generative models like Generative Adversarial Networks (GANs) and Variational Autoencoders (VAEs) as well as privacy-aware methods such as differential privacy that can create synthetic data without revealing any personal information. There have been suggestions of using hybrid techniques that incorporate differential privacy along with either large language models or federated learning to strike a compromise between data confidentiality and availability.

Nevertheless, there have been improvements; still, a lot of obstacles persist. Most synthetic data systems offer a visual realism and scenario diversity but most of them do not have a formal privacy guarantee, which makes these systems open to membership inference and re-identification attacks. Generative models that do not take privacy into consideration may pass on biases present in real datasets to synthetic ones. Moreover, transportation dataset privacy and utility evaluation is hampered by the absence of standard domain-specific metrics, causing heterogeneous performance appraisal and restricted validation frameworks.

7. Brief description of the solution's nature, including a conceptual diagram (maximum 500 words).

The introduced approach is a privacy-preserving synthetic data generation framework meant for the training of autonomous vehicle (AV) models. The system not only fills in the gaps found in earlier studies but also introduces an integration of formal privacy-preserving measures into the synthetic data generation line, thus, treating privacy as a primary concern, not a post-processing step. This procedure reduces the risks of membership inference, re-identification, and sensitive pattern leakage while maintaining the usefulness of data for perception and decision-making models.

The proposed system, unlike the current synthetic data platforms that focus mainly on realism and scalability, enforce privacy guarantees during model calibration and data synthesis. Privacy-preserving methods, for instance, differential privacy-based noise injection and controlled feature abstraction are applied during the learning process from real-world datasets that are used for simulator calibration. This ensures that sensitive attributes are not memorized or reproduced in synthetic outputs. Moreover, the framework encompasses the systematic evaluation of privacy-utility trade-offs that address the absence of standardized validation pointed out in previous research.

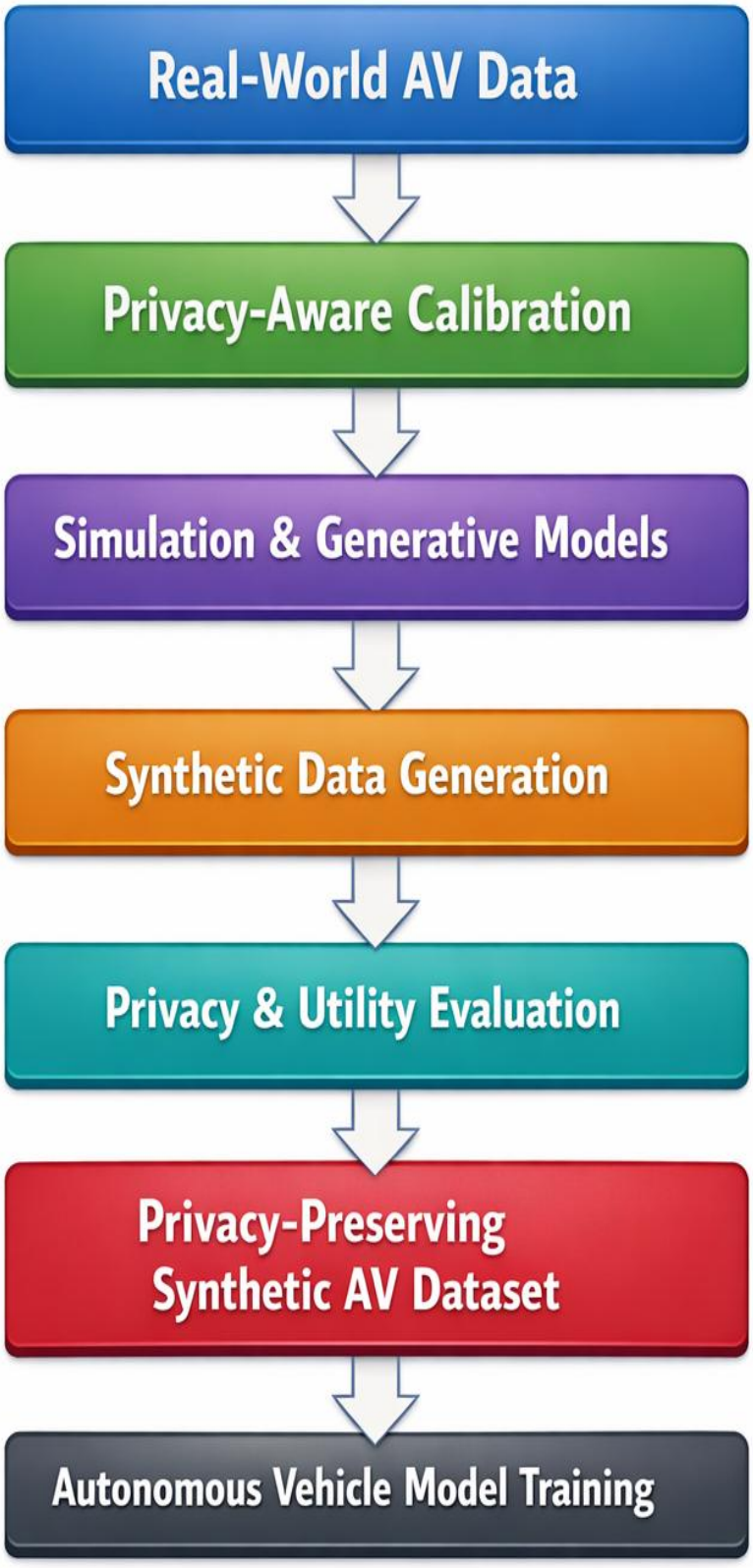
A system-level solution would be as follows:

Input:
- Limited real-world autonomous driving datasets (used only for calibration and benchmarking)
- Simulation parameters (road layouts, traffic density, weather conditions, sensor configurations)
- Privacy parameters (privacy budget, noise levels, risk thresholds)

Process:
- Privacy-aware calibration of simulation and generative models
- Synthetic environment and scenario generation
- Generation of labeled synthetic sensor data (camera, LiDAR, radar)
- Privacy risk evaluation (inference and re-identification testing)
- Utility evaluation using AV perception and prediction models

Output:
- Privacy-preserving synthetic autonomous driving datasets
- Quantitative privacy and utility metrics
- Trained and validated AV models using synthetic data

The solution aligns with the AIMS research cluster by supporting intelligent autonomous systems while embedding Cyber Security principles such as privacy-by-design and secure data handling. It enables scalable, compliant, and secure AV model development without reliance on large volumes of sensitive real-world data.


Real-World AV Data
Privacy-Aware Calibration
Simulation & Generative Models
Synthetic Data Generation
Privacy & Utility Evaluation
Privacy-Preserving Synthetic AV Dataset
Autonomous Vehicle Model Training

8. Brief overview of the availability of necessary specialized domain expertise, knowledge, and data. (500 words max)

The execution of this project successfully relies on multidiscipline expertise that covers autonomous systems, machine learning, synthetic data generation, and cyber security. The necessary domain knowledge can be obtained through a structured review of academic literature about perception systems of autonomous vehicles, simulation-based training environments, and privacy-preserving data techniques. Peer-reviewed journals, conference proceedings (e.g., NeurIPS, CVPR, CoRL), and technical documentation of popular AV simulators are the most trusted learning resources. Furthermore, guidance from academic supervisors in the AIMS research cluster is another way to ensure the alignment of the domain and the rigor of research.

The necessary tech knowledge and personal traits for the project are machine learning, computer vision, generative modeling, and privacy-enhancing technologies. These abilities can be gained by experimentation on the ground using the universally accepted model development frameworks like PyTorch or TensorFlow and synthetic data generation platforms like CARLA or AirSim. Cyber security-specific skills like differential privacy concepts, threat modeling, and privacy risk evaluation (like inference and re-identification attacks) can be acquired from a focused course, online technical modules, and hands-on learning. Version control systems and collaborative development practices facilitate efficient teamwork and reproducibility.

The project will make use of simulation-based synthetic data generation primarily as the main method for data collection, thereby significantly reducing the reliance on sensitive real-world datasets. The usage of publicly available autonomous driving datasets will be done in a limited manner, only for the purposes of simulator calibration, benchmarking, and validation, to mention a few. These datasets are concurrently highly utilized in academic research and are predominantly accessed via predetermined usage licenses. The synthetic data produced in the simulated environments will constitute the primary dataset for the training and evaluation of self-driving car models.

When we talk about the ethical aspects of the project, it is the use of synthetic data that minimizes the ethical risk. Nonetheless, if in any way real-world datasets with personal or location-based data are used for calibration or evaluation, then still ethical clearance will be necessary. In such instances, the project follows the concept of privacy-by-design, guarantees that it is adhering to the relevant data protection laws, and does not allow the identification of individuals through the data being used. The project can count on the support of domain experts, easy access to learning resources, and ethically compliant data sources to be carried out within the academic and regulatory framework.

9. Objectives and Novelty

Main Objective To design, implement, and evaluate a privacy-enhanced synthetic data generation framework that enables secure and compliant training of autonomous vehicle machine learning models while preserving data utility and mitigating privacy leakage risks.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
IT22309556	To integrate formal privacy guarantees into synthetic data generation.	Select privacy models, implement privacy controls, test privacy leakage.	Privacy enforced during data generation rather than post-processing.
IT22541284	To generate realistic, privacy-aware synthetic autonomous vehicle datasets.	Configure simulator, generate labeled data, apply privacy constraints.	Simulation realism combined with explicit privacy parameters.
IT22110220	To evaluate the utility of privacy-preserving synthetic data.	Train AV models, compare performance, analyze privacy utility trade-offs.	Quantified impact of privacy guarantees on AV model performance.
IT22066916	To assess privacy risks and regulatory alignment of the system.	Conduct inference attack tests, map system to data protection principles.	Integrated cyber security and compliance evaluation for AV synthetic data.

10. This part is to be filled by the Supervisor and the Co-supervisor of the Project.

a) This research topic possesses a comprehensive scope suitable for a final-year project.

Yes	✓	No	

b) The proposed topic exhibits novelty.

Yes	✓	No	

c) The student group can successfully execute the proposed project.

Yes	✓	No	

d) The proposed sub-objectives reflect the students' areas of specialization.

Yes	✓	No	

e) Any other comments:

	Title	First Name	Last Name	Signature
Supervisor	Prof	Anuradha	Jayakody	
Co-Supervisor				
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				